Simulation of the Motion of Astronomical Bodies

Darragh Morrissey - 40451054

November 4, 2026

Here's the outline for the report regarding the layout and process.

- Structured work:
  - Structured(1) - Stationary Sun and moving Earth
  - Structured(2) - Track period of Earth's orbit
  - Structured(3) - Track period of new planet based off student id number
- Allow Sun to move due to pull of Earth
- Extend simulation to include all planets of the Solar System
- Allow planets of Solar System to move clockwise and counterclockwise to view how stability is affected
- Add asteroids to simulation to examine bahviour of kirkwood gaps

1 Abstract

Using python, a simulation of the bodies of the Solar System was created

2 Introduction

The aims of this project was to accurately model the behaviour of astronomical bodies in the Solar System and to examine the nature of Kirkwood gaps using python. It was assumed that all motion will take place on the x-y plane

3 Methods

Usage of Runge Kutta method
Sun and Earth model with stationary Sun
Orbital time tracker

Sun and Earth model with Sun able to move

Sun and arbitrary body model where initial velocity is given to body but not Sun.

Give Sun initial momemtum to give system 0 total momentum

Add rest of bodies from Solar System into simulation

Examine Kirkwood gaps by only considering Sun, Jupiter and asteroids at various starting orbits

4 Results

5 Conclusion

6 Appendix